# Acid–Base Titration Curves

Answer key with worked solutions

## A. Understanding

**1.** Equivalence point vs end point.

> The **equivalence point** is where the moles of base added exactly match the moles of acid (from the balanced equation). The **end point** is where the indicator changes colour. A good indicator makes them almost the same.

**2.** Equivalence-point pH.

> (a) **Equal to 7**: strong acid + strong base. (b) **Above 7**: weak acid + strong base (formate is weakly basic). (c) **Below 7**: strong acid + weak base ($NH_4^+$ is weakly acidic).

## B. Strong acid with strong base

**3.** Equivalence volume.

> Moles HCl = 0.150 mol/L × 0.02000 L = 0.00300 mol. Volume NaOH = 0.00300 mol ÷ 0.100 mol/L = 0.0300 L = **30.0 mL**

**4.** Initial pH.

> pH = −log(0.150) = **0.82**

**5.** After 10.00 mL NaOH.

> NaOH added = 0.100 mol/L × 0.01000 L = 0.00100 mol; HCl left = 0.00300 mol − 0.00100 mol = 0.00200 mol in 30.00 mL. [$H_3O^+$] = 0.00200 mol ÷ 0.03000 L = 0.0667 M; pH = **1.18**

**6.** After 40.00 mL NaOH.

> NaOH added = 0.100 mol/L × 0.04000 L = 0.00400 mol; excess $OH^-$ = 0.00400 mol − 0.00300 mol = 0.00100 mol in 60.00 mL. [$OH^-$] = 0.00100 mol ÷ 0.06000 L = 0.0167 M; pOH = 1.78; pH = 14.00 − 1.78 = **12.22**

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## C. Weak acids and indicators

**7.** pKa and Ka from the half-equivalence point.

> At half-equivalence, pH = pKa, so **pKa = 4.20** and $K_a = 10^{-4.20} = \mathbf{6.3 \times 10^{-5}}$

**8.** Indicator choice.

> **Phenolphthalein** (pH 8.2–10.0). The equivalence point of a weak acid–strong base titration is above 7, inside the steep jump where phenolphthalein changes colour. Methyl orange would change far too early.

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